



CYBERSHIELD SANDBOX

GenAI-Powered Social Engineering Awareness Training Platform

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Platform Overview

CyberShield Sandbox trains employees to recognize social engineering attacks – before they happen.



3

Attack Types

Vishing · Smishing · Pretexting



6+

Departments

IT · HR · Finance · Legal · Ops



7

MITRE Techniques

T1566 · T1598 · T1204 · T1592



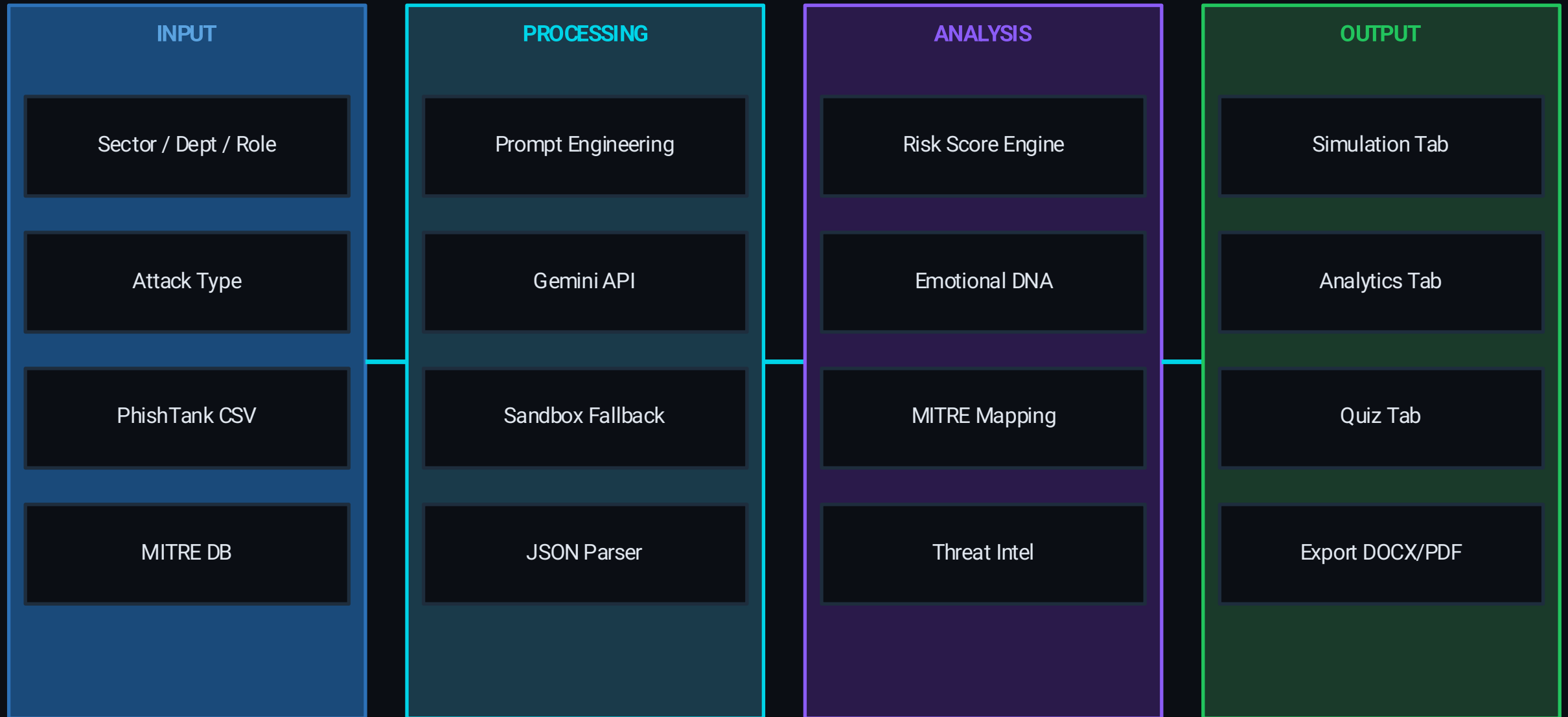
0– 100

Risk Score Range

Composite 3-dimension score

Fully labelled [TRAINING SIMULATION ONLY] · No real credentials harvested · No phishing infrastructure deployed

System Architecture — 4-Layer Design



Simulation Engine

User Config

Sector / Dept / Role / Attack Type

Prompt Engineering

Inject params + safety constraints

Gemini API Call

gemini-1.5-flash · External LLM

JSON Parser

Validate schema · Strip fences

Simulation Object

Full enriched JSON passed downstream

OUTPUT SCHEMA

simulation_text:	"[TRAINING SIMULATION ONLY]..."
manipulation_techniques:	[{technique, snippet, danger}]
emotional_profile:	{fear, urgency, authority...}
threat_intelligence:	{risk_score, likelihood, impact}
mitre_mapping:	[{T1566.003, T1598, ...}]
awareness_card:	{red_flags, what_to_do...}
quiz:	{mcqs[], scenario_question}

⚡ Sandbox fallback: pre-seeded IT Helpdesk + HR scenarios (offline mode)

Threat Analysis Engine

 RISK SCORE (0– 100)

Sophistication of Pretext0–40 pts (40%)



Emotional Cue Density0–30 pts (30%)



Technical Actions Requested0–30 pts (30%)



LIKELIHOOD THRESHOLDS



 MITRE ATT&CK MAPPINGS

- T1566.003

Spearphishing Voice
Vishing
- T1566.002

Spearphishing Link
Smishing
- T1598.003

Phish Info: Voice
Vishing
- T1598.002

Phish Info: Link
Smishing
- T1204.001

User Execution: Malicious
All
- T1592

Gather Victim Host Info
Recon
- T1593

Gather Victim Identity
Recon

Emotional Profile Engine – Attack DNA

Every simulation generates a 7-axis psychological fingerprint that quantifies how the attack exploits human cognition.



Scores visualized as a Plotly radar chart in the Analytics tab, helping trainers identify the dominant psychological attack vectors.

Interactive Training Interface

01



HOME

Pre-seeded quick demo scenarios

PhishTank trend statistics

Threat statistics dashboard

02



SIMULATION

[TRAINING ONLY] banner

Inline manipulation highlights

Red flags + Coach tip boxes

03



ANALYTICS

Risk Score gauge (0–100)

Emotional DNA radar chart

MITRE technique bar chart

04



QUIZ

MCQ × 3–4 options/question

Scenario-based questions

Real-time scoring + explanations

05



EXPORT

DOCX: full training report

PDF: formatted summary

io.BytesIO browser download

Report Export Engine

DOCX (python-docx)

- ✓ Header + Safety Tag
- ✓ Organisation Profile
- ✓ Simulation Scenario Text
- ✓ Threat Intelligence Matrix
- ✓ Emotional DNA Profile
- ✓ Manipulation Techniques
- ✓ MITRE ATT&CK Mappings
- ✓ Quiz Questions & Answers

PDF (ReportLab)

SimpleDocTemplate

Letter size · Paginated

Threat Tables

Metrics formatted in tables

Emotional Profile

7-axis scores with labels

MITRE Maps

Technique IDs + mitigations

Quiz Summary

Q&A with explanations

BytesIO Stream

Browser download button

Safety & Compliance

[TRAINING SIMULATION ONLY] — All simulations carry this label at generation time

✓ PLATFORM DOES

- ✓ Generates defensive awareness simulations
- ✓ Labels ALL content [TRAINING SIMULATION ONLY]
- ✓ Maps to MITRE ATT&CK for evidence-based defence
- ✓ Aligns with NIST CSF & ISO/IEC 27001
- ✓ Provides mitigation strategies per technique

✗ PLATFORM DOES NOT

- ⚠ Does NOT generate real exploit code
- ⚠ Does NOT deploy phishing infrastructure
- ⚠ Does NOT harvest real credentials
- ⚠ Does NOT process personal user data
- ⚠ Does NOT enable real attacks of any kind

System Flow — Sequence & Component Diagram

END-TO-END SEQUENCE (10 Steps)

- 1 User selects sector, dept, role, attack type
- 2 Build structured prompt (gemini_helper.py)
- 3 POST to Gemini API — or load SANDBOX_DATA (fallback)
- 4 JSON simulation response returned from Gemini
- 5 get_mitre_mapping() — technique list + mitigations
- 6 calculate_risk_score() — risk score + emotional DNA
- 7 Enriched simulation object returned to app.py
- 8 Render tabs — Simulation, Analytics, Quiz
- 9 User requests DOCX / PDF export
- 10 BytesIO stream delivered as browser download

COMPONENT FLOW (C1 → C11)

- C1 User Config UI · app.py sidebar → config_dict
- C2 Prompt Engineering · Inject params + safety constraints
- C3 LLM Inference · Gemini API → simulation JSON
- C4 Sandbox Fallback · Pre-seeded offline scenarios
- C5 JSON Parser · Validate + clean LLM output
- C6 MITRE ATT&CK Mapper · Map attack type → technique IDs
- C7 Risk Score Engine · 0–100 composite score
- C8 Emotional Profile Engine · 7-axis Attack DNA scores
- C9 Threat Intel Feed · PhishTank + Emotional CSV stats
- C10 Interactive Training UI · Render simulation + quiz tabs
- C11 Report Exporter · DOCX + PDF via BytesIO stream

Blue border = External API call Green border = MITRE / Analysis Amber border = Threat Intel / Export



CYBERSHIELD SANDBOX

Empowering employees to recognize and resist social engineering — one simulation at a time.

REALISTIC

AI-Generated
Scenarios

MEASURABLE

Risk Scores
& Analytics

SAFE

Defensive
Training Only